

Cryptography sheet # 6

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In this sheet, we will discuss some details of the RSA cryptography system and once again expand our knowledge of rings.

1 Are there enough (good) prime numbers?

It is easy to prove that there are infinitely many prime numbers. Assume that the set $\{p_1, \dots, p_n\}$ contains all prime numbers. Then, you could look at the number $q = p_1 \cdot \dots \cdot p_n + 1$. Using modular arithmetic, it is clear that none of the numbers $p_1, \dots, p_n$ is a divisor of q . This is in contradiction to the fact that every number has a prime factorization.

When we use RSA-cryptography, we want to encrypt using a function $f(x) = x^e$ over $\mathbb{Z}_{p \cdot q}$. For decryption we require that $\gcd(e, p-1) = \gcd(e, q-1) = 1$. Let's say we have already decided which exponent e we want to use. It would be a shame if we could not find some p and q that fulfill the above condition. This raises the following question: How many primes with this property exist? Show that there is an infinite amount of them.

Remark: Some of you might want to solve this exercise by using Dirichlet's theorem on arithmetic progressions. But of course this won't give us any insides. Try to find an elementary proof like the one from the beginning.

2 Exponentiation and the least common divisor

Let a, b be positive integers. The smallest positive integer z such that z is a multiple of a and b (i.e. $z = a \cdot m = b \cdot n$ for some $n, m \in \mathbb{N}$) is called least common multiple of a and b or in short $\text{lcm}(a, b)$. The lcm of a and b can also be written as $\text{lcm}(a, b) = \frac{a \cdot b}{\gcd(a, b)}$. Let ℓ be the least common multiple of $p-1$ and $q-1$. Show that:

$$\forall x \in \mathbb{Z}_{p \cdot q}: \quad x^\ell = [1]_{p \cdot q}$$

What does this imply for the value of $x^{k \bmod \ell}$ with $k \in \mathbb{N}$ and $x \in \mathbb{Z}_{p \cdot q}$.

Hint: the CRT could be useful in this exercise.

3 Variations of RSA decryption

If we perform RSA-cryptography, we choose prime numbers p and q and some $e \in \mathbb{Z}$ with $\gcd(p-1, e) = \gcd(q-1, e) = 1$, and encrypt some $x \in \mathbb{Z}_{p \cdot q}$ as $y = x^e$. There are different approaches for decryption:

- (a) The first one is to compute the inverse d of $[e]_{(p-1) \cdot (q-1)}$ and find y^d .

- (b) The second is to use the CRT to map $y = x^e$ from $\mathbb{Z}_{p \cdot q}$ to $\mathbb{Z}_p \times \mathbb{Z}_q$, and reverse the exponentiation there with the help of Fermat's little theorem. Finally, map the result back to $\mathbb{Z}_{p \cdot q}$ using the EEA.
- (c) The third one is motivated by the exercise above. From there, we see that we might also get the correct result by computing d' as the inverse of $[e]_{\text{lcm}(p-1, q-1)}$ and using this to find $y^{d'}$.

So many options to choose from...but how different are they really? We want to try that out. Use the methods described in (a), (b) and (c) to find x from the data: $e := 3, p := 11, q := 17, y := 36$ and confirm that they all give the same result.

Remark: If you'd like, you can implement the described methods and check if they return the same result for every $y \in \mathbb{Z}_{p \cdot q}$.

4 Rings and non-rings

Are the following structures unitary commutative rings? Justify your answer.

1. The set $\{\text{False}, \text{True}\}$, with xor being the addition and and being the multiplication.
2. The set $\{\text{False}, \text{True}\}$, with or being the addition and and being the multiplication.
3. The set of non-negative integers with a regular addition and multiplication.
4. The set of even integers with a regular addition and multiplication.
5. The set of all functions from $\mathbb{N}$ to $\mathbb{Z}$ with a regular addition and multiplication of functions (i.e. $[f + g](x) = f(x) + g(x)$ and $[f \cdot g](x) = f(x) \cdot g(x)$).

5 Zero divisors

Consider non-zero elements a, b, c in a unitary commutative ring $(R, +, \cdot)$.

1. If $a^2 = 0$, is a a zero divisor?
2. If $a^3 = 0$, is a a zero divisor?
3. If $ab = 0$, is a a zero divisor?
4. If $abc = 0$, is a a zero divisor?
5. If $abc = 0$, is any of the elements a, b, c a zero divisor?
6. If a is a zero divisor, and $ab \neq 0$, is ab a zero divisor?

Justify your answers by giving either an argument why the answer is 'yes' or a counter-example showing that the answer is 'no'.